Mahla Alimardani

Student Researcher · Deep Learning, Statistics & Model Generalization

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https://github.com/mahlaalimardani/MahlaAlimardani

Profile

M.Sc. Artificial Intelligence graduate with a strong statistical foundation (M.Sc. Mathematical Statistics) and hands-on experience training and evaluating deep networks in PyTorch. Particularly interested in how deep networks learn, adapt and generalize across tasks and data distributions - the central questions of continual learning, such as catastrophic forgetting and the stability-plasticity trade-off. Experienced in transfer learning, fine-tuning, controlled experiments and robust evaluation; author of two journal papers on neural networks and machine learning.

Research Projects

Explainable Multimodal Deep Learning for Breast Cancer Classification - M.Sc. thesis

- Designed and implemented a three-stage deep learning pipeline in PyTorch on CBIS-DDSM mammography: U-Net lesion segmentation, EfficientNet-B0 classification and fusion with clinical data.
- Studied how pre-trained networks adapt to a new, small and highly imbalanced domain using transfer learning, fine-tuning and imbalance-aware loss functions.
- Ran systematic experiments on hyperparameters and decision thresholds, reaching a test AUC of 0.898 and 90.5% malignant recall.
- Analyzed model behavior with Grad-CAM++ to show which image regions drive each prediction.
- Protected experimental validity by detecting unreliable labels and rebuilding the image-label mapping from DICOM metadata.

Organoid-to-Tissue Generalization - research project (ongoing)

- Investigating, with machine learning models, whether organoids generalize to the original tissues they are derived from - a question of generalization under distribution shift.
- Building a multi-source dataset by gathering and curating public data from scattered sources for model training and testing.

End-to-End NLP Pipeline

- Built a Python pipeline for text preprocessing, classification and evaluation, improving accuracy through systematic tuning.

Education

M.Sc. Artificial Intelligence - Berlin School of Business & Innovation (BSBI), Germany · 2025 - 2026 (completed, certificate pending)

Thesis: Explainable multimodal deep learning for breast cancer classification using mammography images and clinical data

M.Sc. Mathematical Statistics -Shahrood University of Technology, Iran · 2015 - 2018

B.Sc. Statistics - Gonbad Kavus University, Iran · 2011 - 2015

Publications

- Alimardani, M. and Almasi, M. (2020) 'Investigating the application of particle swarm optimization algorithm in neural networks to increase the accuracy of breast cancer prediction', *International Journal of Computer Trends and Technology*, 68(4), pp. 65-72.
- 'Application of self-organizing map for time-varying biological complex systems prediction', *International Journal of Computer Trends and Technology*, 68(4), pp. 59-64, 2020.

Technical Skills

- **Mathematics & statistics:** probability, statistical inference, linear algebra; experimental design and evaluation
- **Deep learning:** PyTorch, TensorFlow; CNNs (U-Net, EfficientNet), transfer learning, fine-tuning, hyperparameter tuning, diffusion models
- **Python & research software:** object-oriented and functional programming, clean code, design patterns, Git/GitHub, Google Colab (GPU)
- **Data & ML libraries:** NumPy, Pandas, scikit-learn, OpenCV
- **Explainability & LLMs:** Grad-CAM++; OpenAI API, Claude, Hugging Face
- **Other languages & tools:** R, Julia, Java, SQL; VS Code

Professional Experience

Back-End Developer Intern - Hirkan · 06/2022 - 10/2022

- Developed backend components and data pipelines in Java, including socket-based communication.

Data Analyst (Content Production) - Dataak · 11/2020 - 12/2021

- Cleaned and analyzed large banking and social-media datasets with Python and R.

Data Analyst - Sarzamine Talayee Shargh · 01/2018 - 01/2021

- Analyzed product sales data with Python and R to identify trends and support management decisions.

Certificates & Languages

- **Certificates:** Data Analytics (2026) · Python Programming & Practice (2026) · Machine Learning (2020)
- **Languages:** English (fluent) · German (B2) · Persian (native)